



TRAINING PROGRAMME

NLP: from Text Processing to Transformers

Process, represent, classify and extract information from text in English, French and Arabic

27 HOURS

FOR Data scientists and developers who work with text data



Mohamed Aziz BEN HAH

mohamedazizbenhaha.netlify.app

Overview

This programme takes participants through natural language processing from classical text processing to Transformers and large language models. They clean and tokenise text, represent it with bag of words, TF-IDF and word embeddings, build classifiers for sentiment and other tasks, extract entities and relations, and discover topics. They then fine-tune BERT models with Hugging Face, handle multilingual text with attention to Arabic and French, and compare zero- and few-shot LLMs with fine-tuned models. Every chapter ends with a lab, and the programme closes with a final project: an NLP application on multilingual text built and presented by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	Text Processing & Representation	NLP tasks, tokenisation, normalisation, bag of words, TF-IDF, word embeddings	6 h
02	Classical NLP Tasks	Classification and sentiment, named entities and relations, topic modelling	6 h
03	Transformers & LLMs	BERT fine-tuning, multilingual NLP, LLMs for NLP tasks	6 h
04	Final Project	A multilingual NLP application, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	3 h
	Total		27 h

Key facts

AUDIENCE	Data scientists and developers who work with text data
PREREQUISITES	Python and pandas (course: Python for Data Science) and deep learning basics (course: Machine Learning & Deep Learning, deep learning phase), or equivalent. These are used, not taught
DURATION	27 h: 24 h of chapters + 3 h buffer
FORMAT	Short lectures followed by guided labs in Jupyter on English, French and Arabic corpora
LAB ENVIRONMENT	One laptop per participant (16 GB RAM, Python 3, JupyterLab, spaCy, NLTK, Hugging Face); GPU notebooks on Google Colab or a GPU server provided by the organiser for fine-tuning; an open LLM through Ollama or on the GPU server, or a hosted LLM API key per team with a spending cap, provided by the organiser
ASSESSMENT	A lab validated at the end of each phase and a final project presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can build, fine-tune and evaluate NLP models for classification and extraction on English, French and Arabic text, and judge when an LLM is the better tool

PHASE 01

6 H

Text Processing & Representation

Participants see what NLP is used for, then turn raw text into clean tokens with spaCy, NLTK and regular expressions, and into numbers with bag of words, TF-IDF, word2vec and GloVe. Validated lab: a preprocessing pipeline for French and English reviews, and the nearest neighbours of chosen words in an embedding space.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Introduction & Text Processing	Tasks and applications, tokenisation, normalisation, stemming and lemmatisation, regular expressions, spaCy and NLTK	Lecture + Lab	3 h
1.2	Text Representation	Bag of words, TF-IDF, word2vec, GloVe	Lecture + Lab	3 h

Classical NLP Tasks

Participants build classical and neural classifiers for sentiment and other labels, extract named entities and relations, and discover the topics of a corpus with LDA and BERTopic. Validated lab: a sentiment classifier evaluated on a held-out set, and the main topics of a news corpus found with LDA.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Classification & Sentiment	Classical and neural classifiers	Lecture + Lab	3 h
2.2	Information Extraction	Named entities, relations	Lecture + Lab	1.5 h
2.3	Topic Modelling	LDA; BERTopic as a preview of Transformer embeddings	Lecture + Lab	1.5 h

PHASE 03

6 H

Transformers & LLMs

Participants fine-tune BERT-family models with Hugging Face, handle translation and the specifics of Arabic and French text, then use LLMs for zero- and few-shot classification and extraction and compare them with the fine-tuned models. Validated lab: the same classification task solved by a fine-tuned multilingual model and by an LLM, with accuracy and cost compared.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Transformers for NLP	BERT, fine-tuning with Hugging Face	Lecture + Lab	3 h
3.2	Multilingual NLP	Translation, Arabic and French specifics	Lecture + Lab	1.5 h
3.3	LLMs for NLP Tasks	Zero- and few-shot classification and extraction, evaluation against fine-tuned models	Lecture + Lab	1.5 h

PHASE 04

6 H

Final Project

Teams of two or three build an NLP application for classification or extraction on Arabic, French or English text, reusing the phase labs, then present it as they would to a product team.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Brief & Build	Read the brief, prepare the corpus, build a baseline and fine-tune a Transformer model	Project	3 h
4.2	Evaluate & Document	Evaluation per language, error analysis, comparison with an LLM, short report	Project	1.5 h
4.3	Demo & Review	Live demo on new text; questions and feedback	Project	1.5 h

FINAL PROJECT

A Multilingual NLP Application

An application that classifies text or extracts information from Arabic, French or English documents with a fine-tuned Transformer model, evaluated per language and compared with a baseline and an LLM.

STACK Python, spaCy, Hugging Face Transformers and Datasets, scikit-learn, PyTorch, JupyterLab

DELIVERABLE Repository with notebooks and model, evaluation report, live demo